

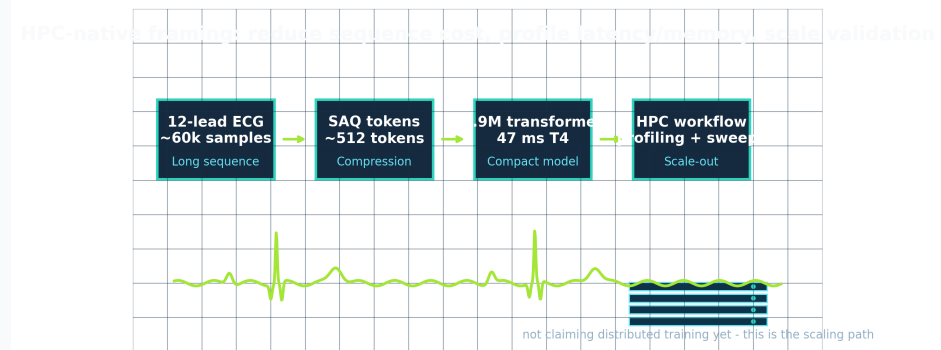
Compute-Efficient Tokenization for Long-Sequence Human Biosignal AI

Segment-Aware Quantisation for compact, calibrated ECG transformers

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HPC-first abstract

ECG is used here as a long-sequence AI testbed. Raw 12-lead records contain tens of thousands of time samples, creating sequence-length, memory, latency and validation bottlenecks. Segment-Aware Quantisation (SAQ) converts cardiac morphology into compact tokens, then feeds a 6.9M-parameter transformer with calibration and constrained thresholds. The computational claim is targeted: preserve diagnosis-critical morphology while reducing modelling cost and opening a path to GPU profiling, distributed sweeps and scalable external validation.



Visual thesis: biosignal sequence -> compact tokens -> lightweight model -> HPC scaling path.

Key repositioning: compute-efficient representation learning for structured physiological sequences - not just ECG classification.

~60k

raw samples
12-lead, 10 sec

~512

SAQ tokens
fixed sequence

6.9M

params
26-28 MB fp32

47 ms

T4 latency
end-to-end

0.959

AUROC
PTB-XL test

0.851

CPSC AUROC
frozen transfer

Retrospective experimental findings only. No patient-care claim. Clinical evaluation remains required.

Sequence length

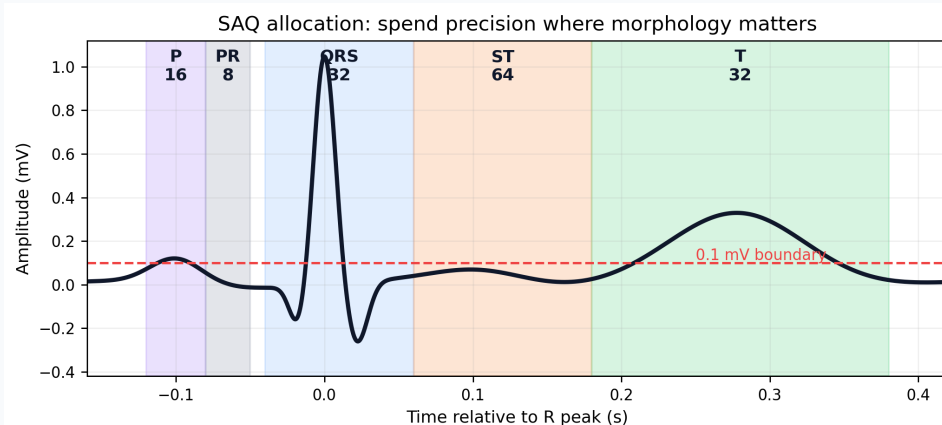
12-lead ECG at 500 Hz can mean ~60,000 values per 10-second record. Modelling raw waveform length directly increases memory and attention/feature cost.

Representation budget

Generic patching can spend capacity on low-information regions while under-resolving morphology that actually matters.

System constraints

The model must report latency, memory, calibration and robustness - not only a headline accuracy number.

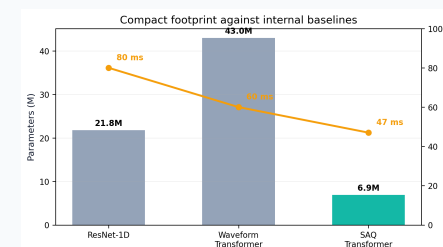


SAQ assigns quantisation levels by cardiac segment importance, then downstrides to a compact token sequence.

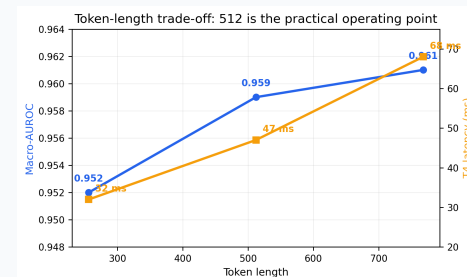
58-75%
sequence reduction
range

0.082 mV
ST RMSE
below 0.1 mV

380 MB
runtime memory
reported



Compute footprint: parameters and T4 latency against controlled baselines.



Token length trade-off: choose 512 because it balances accuracy and latency.



Representation design

- Precision allocation: ST 64, QRS/T 32, P 16, PR 8.
- Cardiac-cycle windows encode physiology before the transformer sees the sequence.
- Novelty: clinical/structural allocation of token precision, not plain uniform quantisation.

Training + calibration

- Three-phase schedule: warm-up, focal-loss main training, hard-negative mining.
- Temperature scaling reduces ECE from 0.433 to 0.127.
- Validation-only thresholds; STEMI threshold enforces sensitivity $\geq 95\%$.

What not to overclaim

- No claim of prospective clinical readiness.
- No claim of MPI/OpenMP implementation yet.
- No claim of external clinical validation beyond frozen CPSC transfer.

Research claim that survives scrutiny

SAQ creates an explicit inductive bias for long physiological sequences, then tests whether compact tokens preserve discrimination, calibration, robustness and efficiency.

4 Evidence: Performance + Robustness

Results are useful only if they survive stress tests

0.959

Macro-AUROC
PTB-XL test

0.921

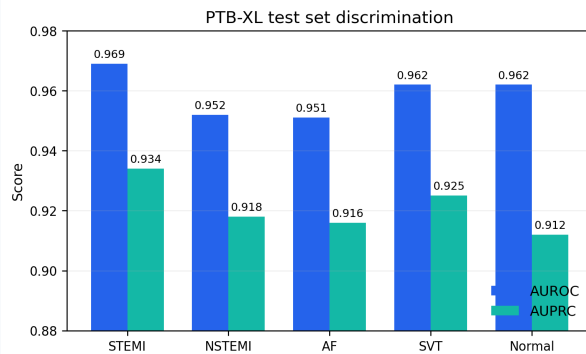
Macro-AUPRC
PTB-XL test

95.2%

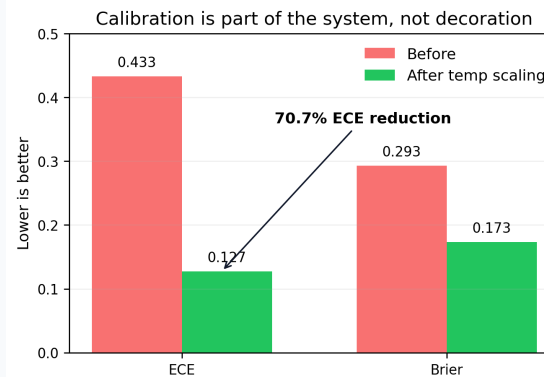
STEMI sensitivity
constrained threshold

0.127

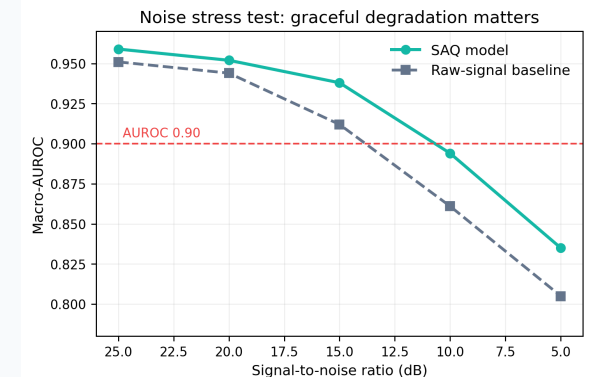
ECE after TS
calibrated output



Per-class discrimination on the PTB-XL held-out test set.



Calibration improves probability quality while discrimination is unchanged.



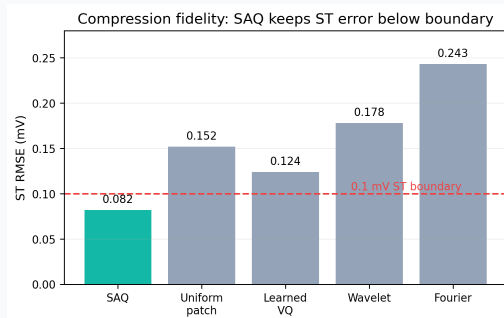
Noise stress-testing shows the SAQ model degrades more gracefully than a raw-signal baseline.

Why this is stronger than a normal ML poster

The evidence is a stack: representation compression, compact model footprint, calibration, constrained thresholds, external frozen transfer and noise injection. That is more defensible than a single accuracy number.

Caveat kept visible

These are retrospective results. The poster should not imply clinical deployment. The IHPCCS angle is computational: efficient modelling and scalable validation for long biomedical sequences.



Compression-method comparison: only SAQ keeps ST RMSE below the 0.1 mV boundary in the matched comparison.

Why this belongs at IHCSS

- Long-sequence AI: ECG modelling is a sequence compression and representation-learning problem.
- Performance engineering: token length, memory footprint and latency directly affect feasibility.
- Scalable validation: larger datasets require reproducible workflows, sweeps and external testing.
- HPC bridge: next step is profiling, workflow automation and multi-dataset experiments on HPC resources.

Next technical steps for IHCSS mentors

GPU profiling

Benchmark kernels, memory, batch-size scaling and end-to-end latency across T4/V100/HPC nodes.

Distributed HPO

Run controlled hyperparameter and ablation sweeps with reproducible manifests.

Workflow packaging

Containerise preprocessing, training, calibration and evaluation for repeatable execution.

External validation

Scale to broader ECG datasets, acquisition devices, noise profiles and subgroups.

Limitations retrospective design; custom five-class taxonomy; fixed segmentation windows; no clinical deployment; HPC scaling is the next work, not a finished claim.

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